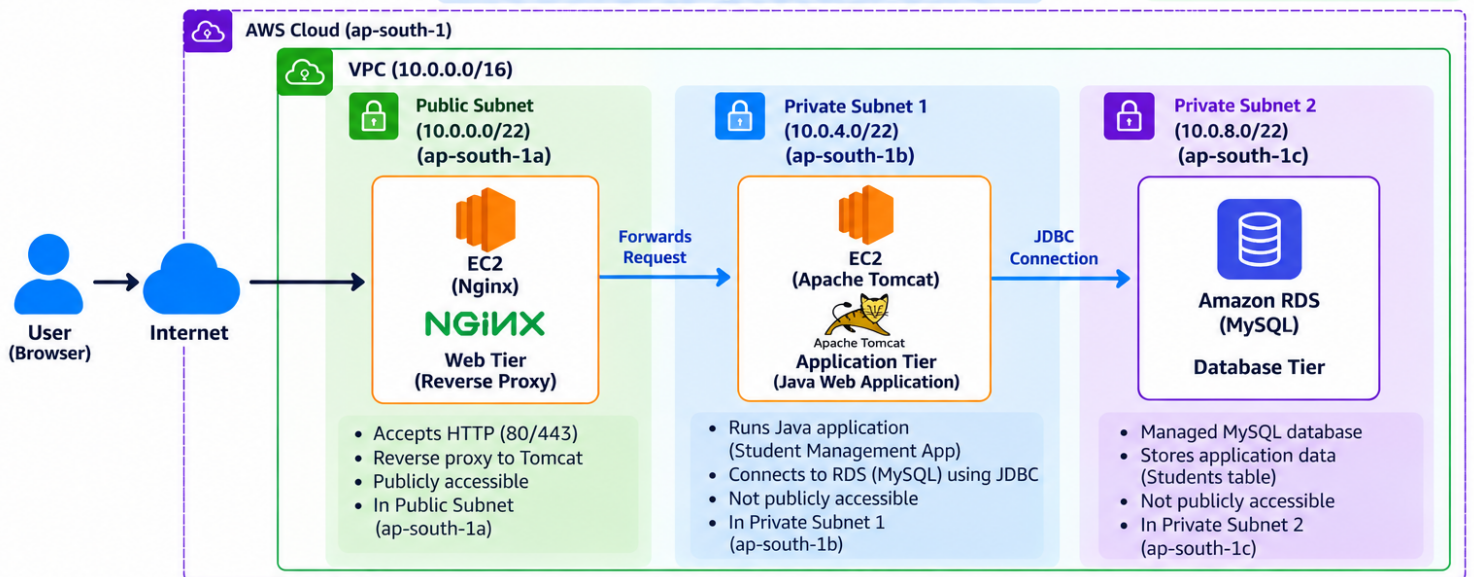




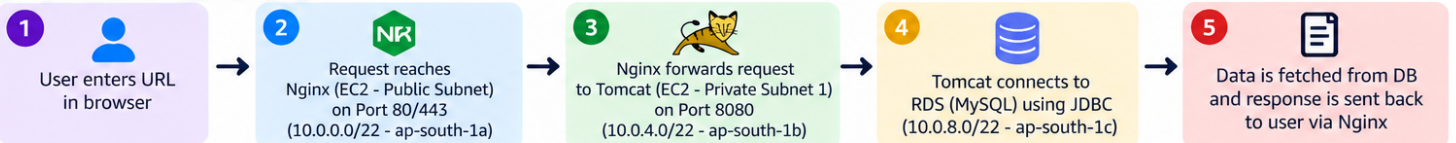
3-Tier Java Application on AWS (Nginx + Apache Tomcat + RDS)

VPC: 10.0.0.0/16 | Region: ap-south-1 (Mumbai)

- ✓ 3-Tier Architecture
- ✓ Secure and Isolated Network
- ✓ Uses RDS (Managed Database)
- ✓ Scalable & Production Ready



Request Flow (Sequence)



Result: User can access the Java web application securely over the internet with the above AWS architecture.

(VPC: 10.0.0.0/16, Public Subnet: 10.0.0.0/22 - ap-south-1a, Private Subnet 1: 10.0.4.0/22 - ap-south-1b, Private Subnet 2: 10.0.8.0/22 - ap-south-1c)

PROJECT INDEX

3-Tier-Project-Java | 26 Project Steps

STEPS 01–13

- | | | |
|----|---|--|
| 01 | VPC Resource Map | VPC architecture and AWS networking resources used for the three-tier |
| 02 | Subnets, CIDR & Availability Zones | Three subnets with their CIDR ranges and Availability Zones. |
| 03 | Three-Tier Security Group | Security group rules used by the project tiers. |
| 04 | Proxy Server EC2 Instance | EC2 instance details for the public proxy/web tier. |
| 05 | Application Server EC2 Instance | EC2 instance details for the private application tier. |
| 06 | Database Server EC2 Instance | EC2 instance details for the private database tier. |
| 07 | SSH Agent & Single Key | SSH agent configured with the single project key for server access. |
| 08 | Hostname & Server Jump | Server hostnames and jump access across the three-tier environment. |
| 09 | Nginx Status, Start & Enable | Nginx service installation/status and service management. |
| 10 | Nginx Reverse Proxy Configuration | Nginx reverse proxy configuration forwarding requests to the application |
| 11 | Java & Apache Tomcat Download | Java verification and Apache Tomcat 9 download/extraction on the application |
| 12 | Tomcat Start | Apache Tomcat started on the application server. |
| 13 | Tomcat Default Page | Tomcat default page verified through the browser. |

STEPS 14–26

- | | | |
|----|--|--|
| 14 | Student WAR Download & Deployment | Student WAR application downloaded and deployed in Tomcat. |
| 15 | Nginx Student Route Configuration | Nginx configured with the /student/ application route. |
| 16 | Student Registration Form | Student registration form displayed through the proxy. |
| 17 | Filled Student Form Before Submit | Student form filled before submitting the request. |
| 18 | Database Connection Error — Troubleshooting | Initial database connection failure captured before adding the JDBC |
| 19 | Amazon RDS Created | Amazon RDS database instance created for the database tier. |
| 20 | MariaDB, RDS Connection & SQL Queries | MariaDB client verification, RDS connection and application SQL queries. |
| 21 | MySQL Connector & Context Configuration | MySQL Connector/J added to the Tomcat library and context.xml opened for |
| 22 | Tomcat Context — RDS Database Configuration | Tomcat context configured with the RDS database connection details. |
| 23 | Final Tomcat & Nginx Restart | Tomcat and Nginx restarted after database configuration. |
| 24 | Final Student Form | Final student registration form displayed. |
| 25 | Successful Student List Output | Successful application output showing the submitted student data. |
| 26 | Database Data Verification | Database data verified after successful application submission. |

Step 1: VPC Resource Map

VPC architecture and AWS networking resources used for the three-tier application.

Subnets (3) Info

Find subnets by attribute or tag

Subnet ID : subnet-082ff8a9d495ae6d9 X Subnet ID : subnet-0a4c4525240d392a6 X Subnet ID : subnet-012645baadb03803 X Show more (+1)

Clear filters

Name	Subnet ID	State	VPC	Block...	IPv4 CIDR	IPv6 ...	IPv6 ...	Avail...	Availability Zone
Public-Subnet	subnet-082ff8...	Available	vpc-0ce5bf849...	Off	10.0.0.0/22	-	-	1019	aps1-az1 (ap-south-1a)
Private-Subnet-2	subnet-01264...	Available	vpc-0ce5bf849...	Off	10.0.8.0/22	-	-	1019	aps1-az2 (ap-south-1c)
Private-Subnet-1	subnet-0a4c45...	Available	vpc-0ce5bf849...	Off	10.0.4.0/22	-	-	1019	aps1-az3 (ap-south-1b)

Step 2: Subnets, CIDR & Availability Zones

Three subnets with their CIDR ranges and Availability Zones.

Your VPCs

VPCs VPC encryption controls

Your VPCs (1/1) Info

Find VPCs by attribute or tag

VPC ID : vpc-0ce5bf849cbac80c3 X Clear filters

Name	VPC ID	State	Encr...	Encr...	Block Public...	IPv4 CIDR	IPv6 CIDR
Three-tier-VPC	vpc-0ce5bf849cbac80c3	Available	-	-	Off	10.0.0.0/16	-

vpc-0ce5bf849cbac80c3 / Three-tier-VPC

Subnets within this VPC

- ap-south-1a Public-Subnet
- ap-south-1b Private-Subnet-1
- ap-south-1c Private-Subnet-2

Route network traffic to resources

- Public-Route-Table
- Private-Route-Table

Connections to other networks

- My-Internet-Gateway
- My-Nat-Gateway

Step 3: Three-Tier Security Group

Security group rules used by the project tiers.

The screenshot displays the AWS Management Console interface for the 'sg-0be2a3c9f2092b555 - Three-Tier-Project-Security-Group'. The left sidebar shows the navigation menu with categories like EC2, Images, Elastic Block Store, and Network & Security. The main content area is titled 'sg-0be2a3c9f2092b555 - Three-Tier-Project-Security-Group' and includes an 'Actions' button. Below the title, there's a 'Details' section with key information: Security group name (Three-Tier-Project-Security-Group), Security group ID (sg-0be2a3c9f2092b555), Description (this group for 3 tier project), VPC ID (vpc-0b51a56429da63957), Owner (026777907119), Inbound rules count (4 Permission entries), and Outbound rules count (1 Permission entry). Below the details, there are tabs for 'Inbound rules', 'Outbound rules', 'Sharing', 'VPC associations', 'Related resources - new', and 'Tags'. The 'Inbound rules' tab is active, showing a table with 4 rules. The table has columns for Name, Security group rule ID, IP version, Type, Protocol, and Port range. The rules are: 'for tomcat' (Custom TCP, port 8080), 'SSH' (SSH, port 22), 'HTTP' (HTTP, port 80), and 'MYSQL' (MYSQL/Aurora, port 3306).

Name	Security group rule ID	IP version	Type	Protocol	Port range
for tomcat	sgr-0272ecfb38d0c2337	IPv4	Custom TCP	TCP	8080
SSH	sgr-0fc505791ba0d4d0d	IPv4	SSH	TCP	22
HTTP	sgr-00fe2959a2c078f8c	IPv4	HTTP	TCP	80
MYSQL	sgr-05986189ce42ca4f8	IPv4	MYSQL/Aurora	TCP	3306

Step 4: Proxy Server EC2 Instance

EC2 instance details for the public proxy/web tier.

The screenshot displays the AWS Management Console interface for the 'i-0992169586393c61f (Proxy-Server)' EC2 instance. The left sidebar shows the navigation menu with categories like EC2, Images, Elastic Block Store, and Network & Security. The main content area is titled 'Instances (1/3) Info' and includes buttons for 'Connect', 'Instance state', 'Actions', and 'Launch instances'. Below the title, there's a table listing the instances. The table has columns for Name, Instance ID, Instance state, Instance type, Status check, Availability Zone, Public IPv4 DNS, and Private IP. The instances are: 'Proxy-Server' (Running, t3.micro, 3/3 checks passed, ap-south-1a, 13.207.41.4), 'Database-Server' (Running, t3.micro, 3/3 checks passed, ap-south-1c, 10.0.1.244), and 'Application-Server' (Running, t3.micro, 3/3 checks passed, ap-south-1b, 10.0.1.244). Below the table, there's a detailed view of the 'i-0992169586393c61f (Proxy-Server)' instance. The details include: Instance ID (i-0992169586393c61f), IPv6 address (None), Hostname type (IP name: ip-10-0-1-244.ap-south-1.compute.internal), Answer private resource DNS name (None), Auto-assigned IP address (13.207.41.4 [Public IP]), Public IPv4 address (13.207.41.4 | open address), Instance state (Running), Private IP DNS name (IPV4 only) (ip-10-0-1-244.ap-south-1.compute.internal), Instance type (t3.micro), VPC ID (vpc-0ce5bf849cbac80c3 (Three-tier-VPC)), Private IPv4 addresses (10.0.1.244), Public DNS (None), Elastic IP addresses (None), and AWS Compute Optimizer finding (Opt-in to AWS Compute Optimizer for recommendations. | Learn more).

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Private IP
Proxy-Server	i-0992169586393c61f	Running	t3.micro	3/3 checks passed	ap-south-1a	13.207.41.4	10.0.1.244
Database-Server	i-08f382844986665e4	Running	t3.micro	3/3 checks passed	ap-south-1c	10.0.1.244	10.0.1.244
Application-Server	i-0fc3ccade9ee865dc	Running	t3.micro	3/3 checks passed	ap-south-1b	10.0.1.244	10.0.1.244

Step 5: Application Server EC2 Instance

EC2 instance details for the private application tier.

The screenshot shows the AWS Management Console interface. On the left sidebar, the 'EC2' menu is expanded, and the 'Instances' link is selected. The main panel displays a list of EC2 instances under the heading 'Instances (1/3) Info'. The list includes three instances: 'Proxy-Server', 'Database-Server', and 'Application-Server'. The 'Application-Server' instance is selected, and its details are shown in the right pane. The details include the instance ID, state (Running), type (t3.micro), status check (3/3 checks passed), availability zone (ap-south-1b), and public IPv4 DNS. The instance is running in a VPC (vpc-0ce5bf849cbac80c3) and has a private IP address of 10.0.5.48.

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Private IPv4 DNS
Proxy-Server	i-0992169586393c61f	Running	t3.micro	3/3 checks passed	ap-south-1a	-	-
Database-Server	i-08f382844986665e4	Running	t3.micro	3/3 checks passed	ap-south-1c	-	-
Application-Server	i-0fc3ccade9ee865dc	Running	t3.micro	3/3 checks passed	ap-south-1b	-	-

i-0fc3ccade9ee865dc (Application-Server)

Instance ID	Public IPv4 address	Private IPv4 addresses
i-0fc3ccade9ee865dc	-	10.0.5.48

Instance state
Running

Private IP DNS name (IPv4 only)
ip-10-0-5-48.ap-south-1.compute.internal

Instance type
t3.micro

VPC ID
vpc-0ce5bf849cbac80c3 (Three-tier-VPC)

Private IP DNS name (IPv4 only)
ip-10-0-5-48.ap-south-1.compute.internal

Answer private resource DNS name
-

Auto-assigned IP address
-

Elastic IP addresses
-

AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations. | Learn more

Step 6: Database Server EC2 Instance

EC2 instance details for the private database tier.

The screenshot shows the AWS Management Console interface. On the left sidebar, the 'EC2' menu is expanded, and the 'Instances' link is selected. The main panel displays a list of EC2 instances under the heading 'Instances (1/3) Info'. The list includes three instances: 'Proxy-Server', 'Database-Server', and 'Application-Server'. The 'Database-Server' instance is selected, and its details are shown in the right pane. The details include the instance ID, state (Running), type (t3.micro), status check (3/3 checks passed), availability zone (ap-south-1c), and public IPv4 DNS. The instance is running in a VPC (vpc-0ce5bf849cbac80c3) and has a private IP address of 10.0.11.249.

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Private IPv4 DNS
Proxy-Server	i-0992169586393c61f	Running	t3.micro	3/3 checks passed	ap-south-1a	-	-
Database-Server	i-08f382844986665e4	Running	t3.micro	3/3 checks passed	ap-south-1c	-	-
Application-Server	i-0fc3ccade9ee865dc	Running	t3.micro	3/3 checks passed	ap-south-1b	-	-

i-08f382844986665e4 (Database-Server)

Instance summary

Instance ID	Public IPv4 address	Private IPv4 addresses
i-08f382844986665e4	-	10.0.11.249

Instance state
Running

Private IP DNS name (IPv4 only)
ip-10-0-11-249.ap-south-1.compute.internal

Instance type
t3.micro

VPC ID
vpc-0ce5bf849cbac80c3 (Three-tier-VPC)

Private IP DNS name (IPv4 only)
ip-10-0-11-249.ap-south-1.compute.internal

Answer private resource DNS name
-

Auto-assigned IP address
-

Elastic IP addresses
-

AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations. | Learn more

SSH agent configured with the single project key for server access.

```
ec2-user@ip-10-0-1-244:~  
praje@Prem_07 MINGW64 ~/Downloads (main)  
$ eval "$(ssh-agent -s)"  
Agent pid 194  
  
praje@Prem_07 MINGW64 ~/Downloads (main)  
$ ssh-add -sh  
Identity added: C:\Users\prajesh\AppData\Local\Microsoft\WindowsApps\OpenSSH-WinSshAgent.exe  
  
praje@Prem_07 MINGW64 ~/Downloads (main)  
$ ssh-add -l  
2048 SHA256:fSVbmE8mSlG9nvnxT0Aszk6Sv6IW6dn9BLedjULWDLw  
  
praje@Prem_07 MINGW64 ~/Downloads (main)  
$ ssh -A -i C:\Users\prajesh\AppData\Local\Microsoft\WindowsApps\OpenSSH-WinSshAgent.exe ec2-user@13.207.41.4  
The authenticity of host '13.207.41.4 (13.207.41.4)' can't be established.  
ED25519 key fingerprint is: SHA256:itVVKXcX5AhONNb9SvgPmmkgCZlsgYs0/c91bek7CZQ  
This key is not known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '13.207.41.4' (ED25519) to the list of known hosts.  
** WARNING: connection is not using a post-quantum key exchange algorithm.  
** This session may be vulnerable to "store now, decrypt later" attacks.  
** The server may need to be upgraded. See https://openssh.com/pq.html  
  
#_  
~\_##### Amazon Linux 2023  
~\_#####\  
~\_####|  
~\_##/  
~\_V~'->  
~\_./.._|  
~\_/_/___|  
~\_/_/___|  
[ec2-user@ip-10-0-1-244 ~]$
```

Server hostnames and jump access across the three-tier environment.

```
ec2-user@database:~
```

```
praje@Prem_07 MINGW64 ~/Downloads (main)
$ ssh -A -i /home/praje/.ssh/id_rsa ec2-user@13.207.41.4
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html

#_
~\##### Amazon Linux 2023
~~\#####\
~~ \###|
~~ \#/      https://aws.amazon.com/linux/amazon-linux-2023
~~ V~'-'>
~~~~
~~~~_. _ ._/ 
    _/_/_/_/_/_/
    _/m/'-'>

Last login: Fri Sep 11 12:46:39 2026 from 152.58.31.124
[ec2-user@proxy ~]$ ssh -A ec2-user@10.0.5.48

#_
~\##### Amazon Linux 2023
~~\#####\
~~ \###|
~~ \#/      https://aws.amazon.com/linux/amazon-linux-2023
~~ V~'-'>
~~~~
~~~~_. _ ._/ 
    _/_/_/_/_/_/
    _/m/'-'>

Last login: Fri Sep 11 12:47:44 2026 from 10.0.1.244
[ec2-user@application ~]$ ssh ec2-user@10.0.11.249

#_
~\##### Amazon Linux 2023
~~\#####\
~~ \###|
~~ \#/      https://aws.amazon.com/linux/amazon-linux-2023
~~ V~'-'>
~~~~
~~~~_. _ ._/ 
    _/_/_/_/_/_/
    _/m/'-'>

Last login: Fri Sep 11 12:49:22 2026 from 10.0.5.48
[ec2-user@database ~]$
```

Step 9: Nginx Status, Start & Enable

Nginx service installation/status and service management.

```
ec2-user@proxy:~  
[ec2-user@proxy ~]$ sudo systemctl status nginx  
● nginx.service - The nginx HTTP and reverse proxy server  
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; disabled; preset: disabled)  
   Active: inactive (dead)  
[ec2-user@proxy ~]$ sudo systemctl enable nginx  
Created symlink /etc/systemd/system/multi-user.target.wants/nginx.service → /usr/lib/systemd/system/nginx.service.  
[ec2-user@proxy ~]$ sudo systemctl start nginx  
[ec2-user@proxy ~]$ sudo systemctl status nginx  
● nginx.service - The nginx HTTP and reverse proxy server  
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: disabled)  
   Active: active (running) since Fri 2026-09-11 12:51:34 UTC; 3s ago  
     Process: 26358 ExecStartPre=/usr/bin/rm -f /run/nginx.pid (code=exited, status=0/SUCCESS)  
     Process: 26359 ExecStartPre=/usr/sbin/nginx -t (code=exited, status=0/SUCCESS)  
     Process: 26360 ExecStart=/usr/sbin/nginx (code=exited, status=0/SUCCESS)  
    Main PID: 26361 (nginx)  
      Tasks: 3 (limit: 1059)  
     Memory: 3.6M  
        CPU: 59ms  
    CGroup: /system.slice/nginx.service  
            └─26361 "nginx: master process /usr/sbin/nginx"  
              └─26362 "nginx: worker process"  
                └─26363 "nginx: worker process"  
  
Sep 11 12:51:34 proxy systemd[1]: Starting nginx.service - The nginx HTTP and reverse proxy server...  
Sep 11 12:51:34 proxy nginx[26359]: nginx: the configuration file /etc/nginx/nginx.conf syntax is ok  
Sep 11 12:51:34 proxy nginx[26359]: nginx: configuration file /etc/nginx/nginx.conf test is successful  
Sep 11 12:51:34 proxy systemd[1]: Started nginx.service - The nginx HTTP and reverse proxy server.  
[ec2-user@proxy ~]$
```

Step 10: Nginx Reverse Proxy Configuration

Nginx reverse proxy configuration forwarding requests to the application server.

```
ec2-user@proxy:/etc/nginx
worker_processes auto;
error_log /var/log/nginx/error.log notice;
pid /run/nginx.pid;

# Load dynamic modules. See /usr/share/doc/nginx/README.dynamic.
include /usr/share/nginx/modules/*.conf;

events {
    worker_connections 1024;
}

http {
    log_format main '$remote_addr - $remote_user [$time_local] "$request" '
        '$status $body_bytes_sent "$http_referer" '
        '"$http_user_agent" "$http_x_forwarded_for"';

    access_log /var/log/nginx/access.log main;

    sendfile on;
    tcp_nopush on;
    keepalive_timeout 65;
    types_hash_max_size 4096;

    include /etc/nginx/mime.types;
    default_type application/octet-stream;

    # Load modular configuration files from the /etc/nginx/conf.d directory.
    # See http://nginx.org/en/docs/nginx_core_module.html#include
    # for more information.
    include /etc/nginx/conf.d/*.conf;

    server {
        listen 80;
        listen [::]:80;
        server_name _;
        root /usr/share/nginx/html;

        # Load configuration files for the default server block.
        include /etc/nginx/default.d/*.conf;

        error_page 404 /404.html;
        location = /404.html {
        }

        error_page 500 502 503 504 /50x.html;
        location = /50x.html {
        }

        location /{
            proxy_pass http://10.0.5.48:8080;
        }

    }

    # Settings for a TLS enabled server.
    #
    -- INSERT --
```

Step 11: Java & Apache Tomcat Download

Java verification and Apache Tomcat 9 download/extraction on the application server.

```
ec2-user@application:~$ java --version
openjdk 26.0.2.1 2026-08-18
OpenJDK Runtime Environment Corretto-26.0.2.11.1 (build 26.0.2.1+11-FR)
OpenJDK 64-Bit Server VM Corretto-26.0.2.11.1 (build 26.0.2.1+11-FR, mixed mode, sharing)
[ec2-user@application ~]$ sudo wget https://dlcdn.apache.org/tomcat/tomcat-9/v9.0.121/bin/apache-tomcat-9.0.121.tar.gz
--2026-09-11 13:23:24-- https://dlcdn.apache.org/tomcat/tomcat-9/v9.0.121/bin/apache-tomcat-9.0.121.tar.gz
Resolving dlcdn.apache.org (dlcdn.apache.org)... 151.101.2.132, 2a04:4e42::644
Connecting to dlcdn.apache.org (dlcdn.apache.org)|151.101.2.132|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 13178703 (13M) [application/x-gzip]
Saving to: 'apache-tomcat-9.0.121.tar.gz'

apache-tomcat-9.0.121.tar.gz      100%[=====] 12.57M  --.-KB/s  in 0.04s

2026-09-11 13:23:24 (317 MB/s) - 'apache-tomcat-9.0.121.tar.gz' saved [13178703/13178703]

[ec2-user@application ~]$ ls
apache-tomcat-9.0.121.tar.gz
[ec2-user@application ~]$ sudo tar -xvzf apache-tomcat-9.0.121.tar.gz -C /opt
apache-tomcat-9.0.121/conf/
apache-tomcat-9.0.121/conf/catalina.policy
apache-tomcat-9.0.121/conf/catalina.properties
apache-tomcat-9.0.121/conf/context.xml
apache-tomcat-9.0.121/conf/jaspic-providers.xml
apache-tomcat-9.0.121/conf/jaspic-providers.xsd
apache-tomcat-9.0.121/conf/logging.properties
apache-tomcat-9.0.121/conf/server.xml
apache-tomcat-9.0.121/conf/tomcat-users.xml
apache-tomcat-9.0.121/conf/tomcat-users.xsd
apache-tomcat-9.0.121/conf/web.xml
apache-tomcat-9.0.121/bin/
apache-tomcat-9.0.121/lib/
apache-tomcat-9.0.121/logs/
apache-tomcat-9.0.121/temp/
apache-tomcat-9.0.121/webapps/
apache-tomcat-9.0.121/webapps/ROOT/
apache-tomcat-9.0.121/webapps/ROOT/WEB-INF/
apache-tomcat-9.0.121/webapps/docs/
apache-tomcat-9.0.121/webapps/docs/META-INF/
apache-tomcat-9.0.121/webapps/docs/WEB-INF/
apache-tomcat-9.0.121/webapps/docs/WEB-INF/jsp/
apache-tomcat-9.0.121/webapps/docs/annotationapi/
apache-tomcat-9.0.121/webapps/docs/api/
apache-tomcat-9.0.121/webapps/docs/appdev/
apache-tomcat-9.0.121/webapps/docs/appdev/sample/
apache-tomcat-9.0.121/webapps/docs/appdev/sample/docs/
apache-tomcat-9.0.121/webapps/docs/appdev/sample/src/
apache-tomcat-9.0.121/webapps/docs/appdev/sample/src/mypackage/
apache-tomcat-9.0.121/webapps/docs/appdev/sample/web/
apache-tomcat-9.0.121/webapps/docs/appdev/sample/web/WEB-INF/
apache-tomcat-9.0.121/webapps/docs/appdev/sample/web/images/
apache-tomcat-9.0.121/webapps/docs/architecture/
apache-tomcat-9.0.121/webapps/docs/architecture/requestProcess/
apache-tomcat-9.0.121/webapps/docs/architecture/startup/
apache-tomcat-9.0.121/webapps/docs/config/
apache-tomcat-9.0.121/webapps/docs/elapi/
apache-tomcat-9.0.121/webapps/docs/images/
apache-tomcat-9.0.121/webapps/docs/images/fonts/
```

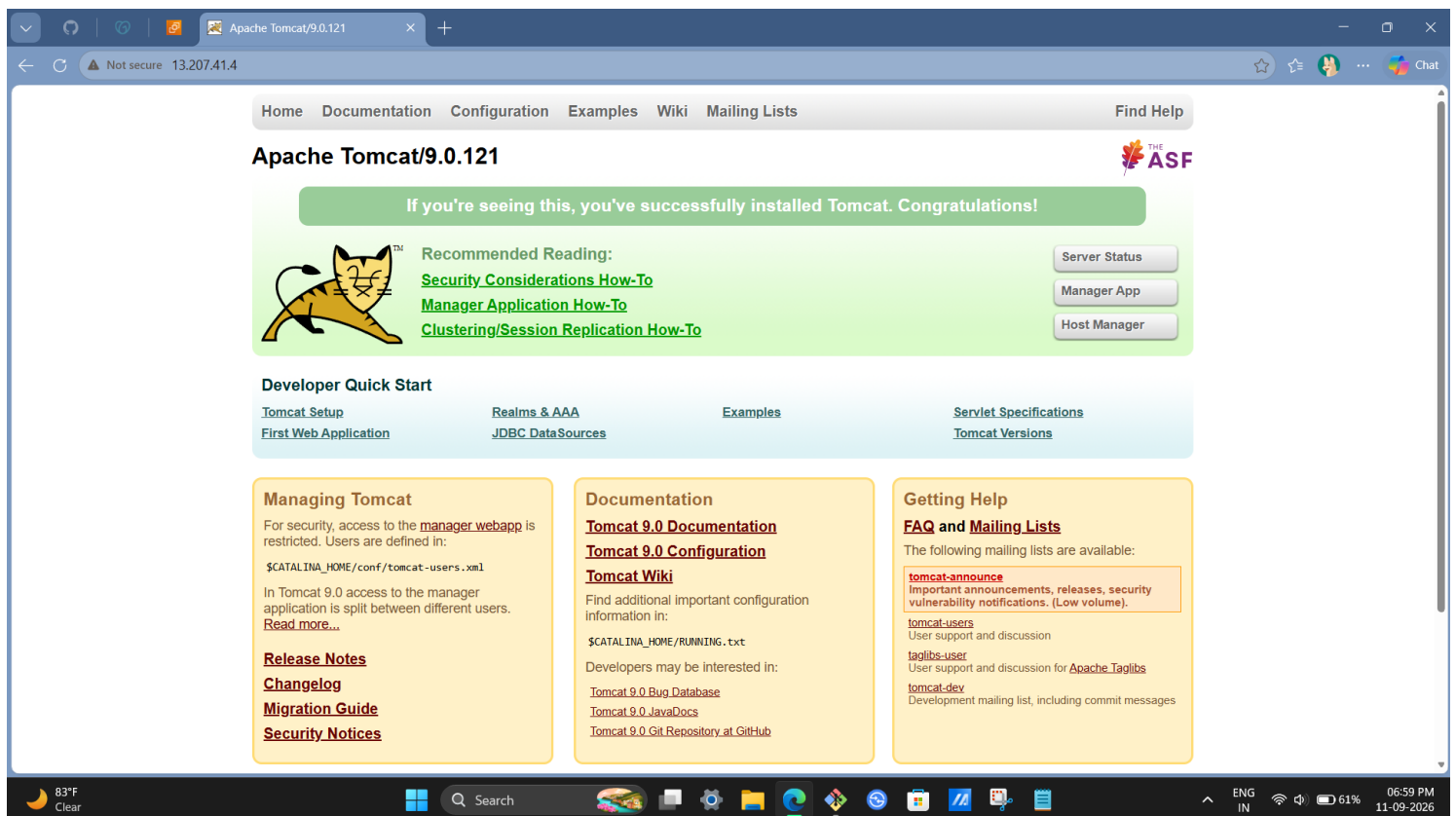
Step 12: Tomcat Start

Apache Tomcat started on the application server.

```
root@application:/opt/tomcat/bin
[ec2-user@application ~]$ sudo -i
[root@application ~]# cd /opt/
[root@application opt]# ls
apache-tomcat-9.0.121 aws
[root@application opt]# mv apache-tomcat-9.0.121 tomcat
[root@application opt]# ls
aws tomcat
[root@application opt]# cd tomcat/bin/
[root@application bin]# ls
bootstrap.jar catalina.sh commons-daemon-native.tar.gz configtest.sh digest.sh setclasspath.bat shutdown.sh tomcat-juli.jar tool-wrapper.sh
catalina-tasks.xml ciphers.bat commons-daemon.jar daemon.sh makebase.bat setclasspath.sh startup.bat tomcat-native.tar.gz version.bat
catalina.bat ciphers.sh configtest.bat digest.bat makebase.sh shutdown.bat startup.sh tool-wrapper.bat version.sh
[root@application bin]# ./catalina.sh start
Using CATALINA_BASE: /opt/tomcat
Using CATALINA_HOME: /opt/tomcat
Using CATALINA_TMPDIR: /opt/tomcat/temp
Using JRE_HOME: /usr
Using CLASSPATH: /opt/tomcat/bin/bootstrap.jar:/opt/tomcat/bin/tomcat-juli.jar
Using CATALINA_OPTS:
Tomcat started.
[root@application bin]# |
```

Step 13: Tomcat Default Page

Tomcat default page verified through the browser.



Step 14: Student WAR Download & Deployment

Student WAR application downloaded and deployed in Tomcat.

```
root@application:opt/tomcat/bin
[ec2-user@application ~]$ sudo -i
[root@application ~]# cd /opt/tomcat/webapps/
[root@application webapps]# ls
ROOT docs examples host-manager manager
[root@application webapps]# curl -L -o student.war https://github.com/Sushant123456789/Three-Tier-War-File/raw/refs/heads/main/student.war
% Total % Received % Xferd Average Speed Time Time Time Current
         Dload Upload Total Spent Left Speed
0 0 0 0 0 0 0 0 --:--:-- --:--:-- --:--:-- 0
100 89423 100 89423 0 0 96988 0 --:--:-- --:--:-- --:--:-- 96988
[root@application webapps]# ls
ROOT docs examples host-manager manager student.war
[root@application webapps]# ls
ROOT docs examples host-manager manager student student.war
[root@application webapps]# cd ../bin/
[root@application bin]# ls
bootstrap.jar catalina.sh commons-daemon-native.tar.gz configtest.sh digest.sh setclasspath.bat shutdown.sh tomcat-juli.jar tool-wrapper.sh
catalina-tasks.xml ciphers.bat commons-daemon.jar daemon.sh makebase.bat setclasspath.sh startup.bat tomcat-native.tar.gz version.bat
catalina.bat ciphers.sh configtest.bat digest.bat makebase.sh shutdown.bat startup.sh tool-wrapper.bat version.sh
[root@application bin]# ./catalina.sh stop
Using CATALINA_BASE: /opt/tomcat
Using CATALINA_HOME: /opt/tomcat
Using CATALINA_TMPDIR: /opt/tomcat/temp
Using JRE_HOME: /usr
Using CLASSPATH: /opt/tomcat/bin/bootstrap.jar:/opt/tomcat/bin/tomcat-juli.jar
NOTE: Picked up JDK_JAVA_OPTIONS: --add-opens=java.base/java.lang=ALL-UNNAMED --add-opens=java.base/java.lang.invoke=ALL-UNNAMED --add-opens=java.base/java.lang.reflect=ALL-UNNAMED --add-opens=java.base/java.i
o=ALL-UNNAMED --add-opens=java.base/java.util=ALL-UNNAMED --add-opens=java.base/java.util.concurrent=ALL-UNNAMED --add-opens=java.rmi/sun.rmi.transport=ALL-UNNAMED
[root@application bin]# ./catalina.sh start
Using CATALINA_BASE: /opt/tomcat
Using CATALINA_HOME: /opt/tomcat
Using CATALINA_TMPDIR: /opt/tomcat/temp
Using JRE_HOME: /usr
Using CLASSPATH: /opt/tomcat/bin/bootstrap.jar:/opt/tomcat/bin/tomcat-juli.jar
Using CATALINA_OPTS:
Tomcat started.
[root@application bin]# |
```

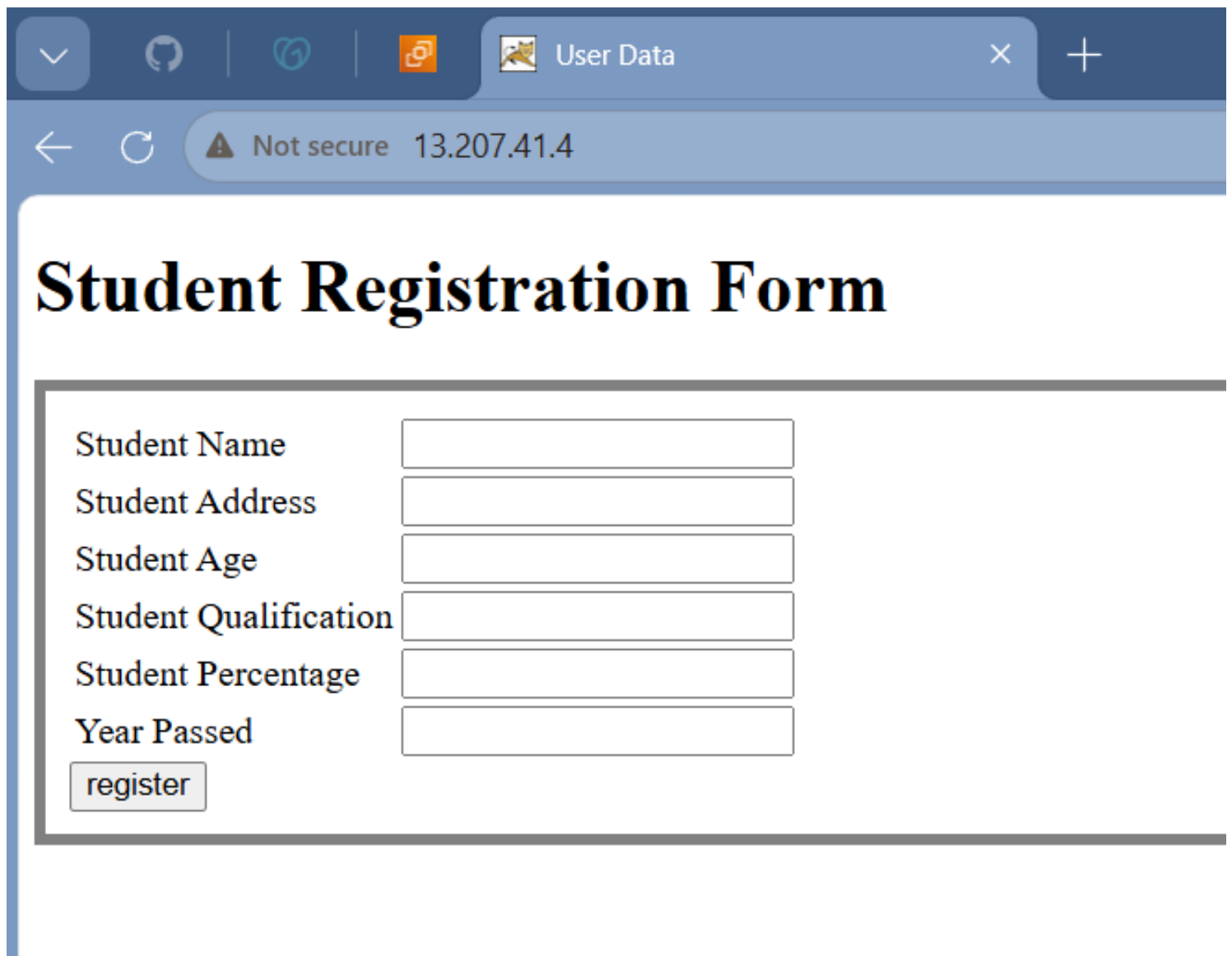
Step 15: Nginx Student Route Configuration

Nginx configured with the /student/ application route.

```
ec2-user@proxy:~  
GNU nano 8.3  
  
http {  
    log_format  main  '$remote_addr - $remote_user [$time_local] "$request" '  
                      '$status $body_bytes_sent "$http_referer" '  
                      '"$http_user_agent" "$http_x_forwarded_for"';  
  
    access_log  /var/log/nginx/access.log  main;  
  
    sendfile            on;  
    tcp_nopush          on;  
    keepalive_timeout   65;  
    types_hash_max_size 4096;  
  
    include             /etc/nginx/mime.types;  
    default_type         application/octet-stream;  
  
    # Load modular configuration files from the /etc/nginx/conf.d directory.  
    # See http://nginx.org/en/docs/nginx_core_module.html#include  
    # for more information.  
    include /etc/nginx/conf.d/*.conf;  
  
    server {  
        listen      80;  
        listen      [::]:80;  
        server_name  _;  
        root         /usr/share/nginx/html;  
  
        # Load configuration files for the default server block.  
        include /etc/nginx/default.d/*.conf;  
  
        error_page 404 /404.html;  
        location = /404.html {  
        }  
  
        error_page 500 502 503 504 /50x.html;  
        location = /50x.html {  
        }  
  
        location /{  
            proxy_pass http://10.0.5.48:8080/student/;  
        }  
  
    }  
}
```

Step 16: Student Registration Form

Student registration form displayed through the proxy.

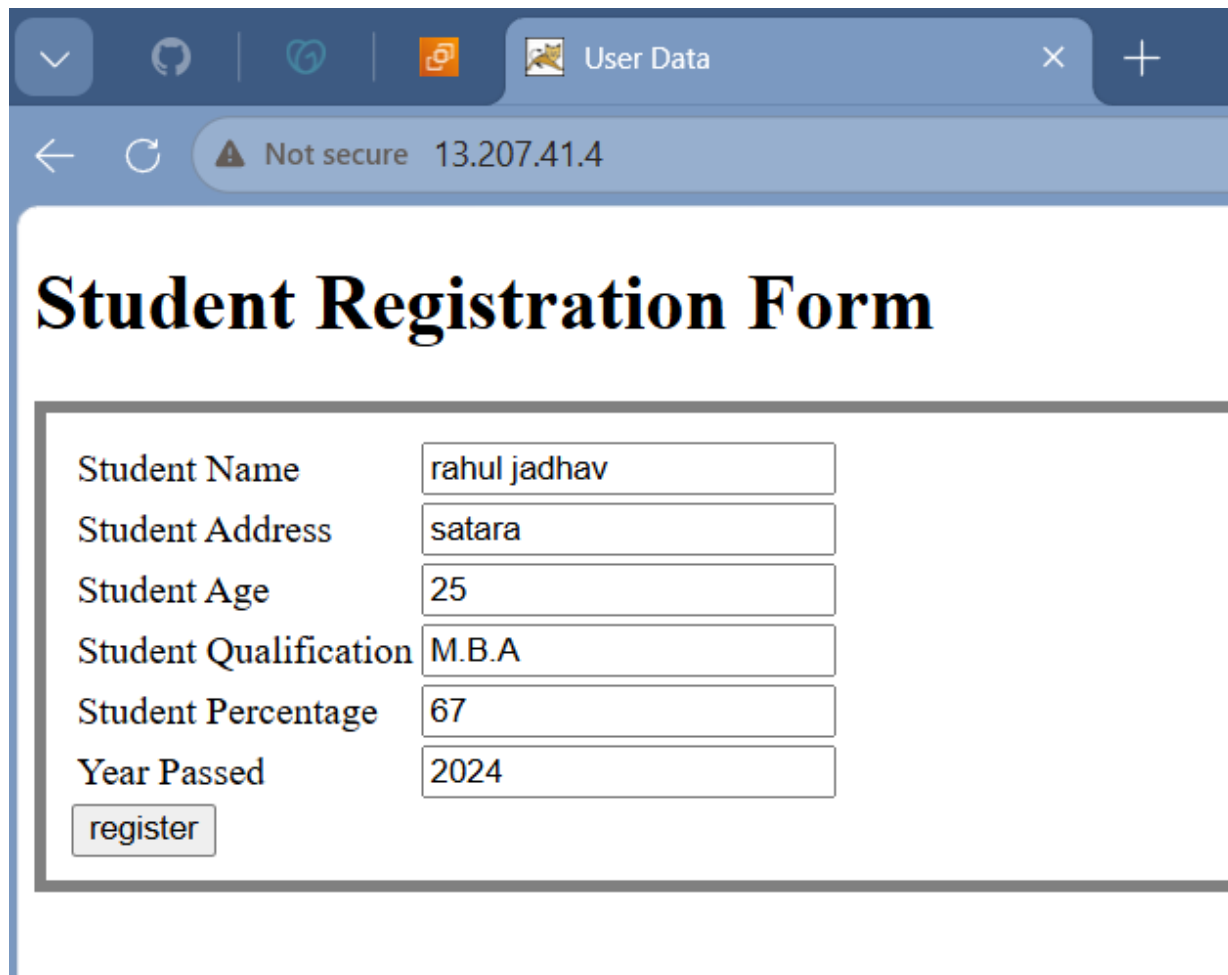


The screenshot shows a web browser window with a single tab titled "User Data". The address bar displays "Not secure" and the IP address "13.207.41.4". The main content area features a large heading "Student Registration Form". Below the heading is a registration form with the following fields and a submit button:

Student Name	
Student Address	
Student Age	
Student Qualification	
Student Percentage	
Year Passed	
register	

Step 17: Filled Student Form Before Submit

Student form filled before submitting the request.



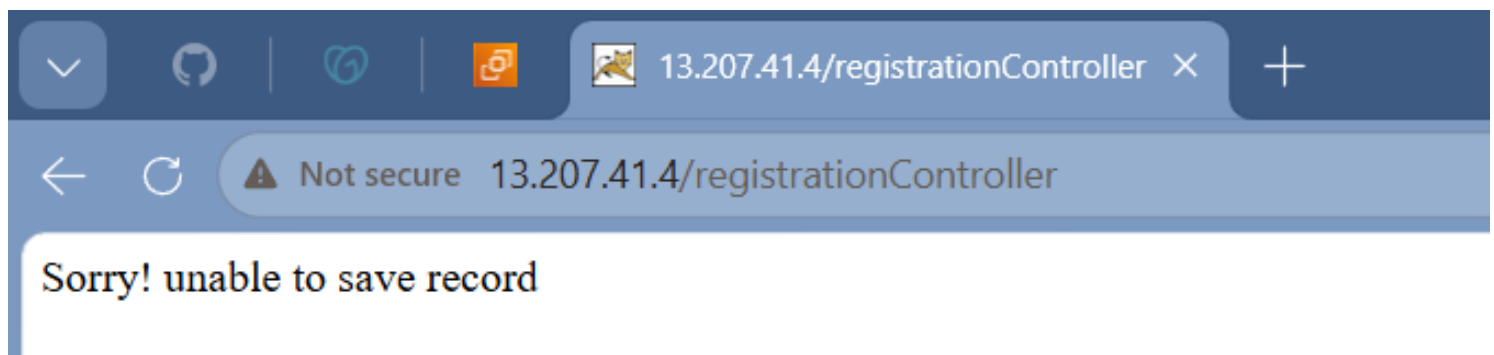
The screenshot shows a web browser window with a single tab titled 'User Data'. The address bar displays '13.207.41.4' with a 'Not secure' warning. The main content area features a large heading 'Student Registration Form'. Below the heading is a form with the following fields and values:

Student Name	rahul jadhav
Student Address	satara
Student Age	25
Student Qualification	M.B.A
Student Percentage	67
Year Passed	2024

At the bottom left of the form is a button labeled 'register'.

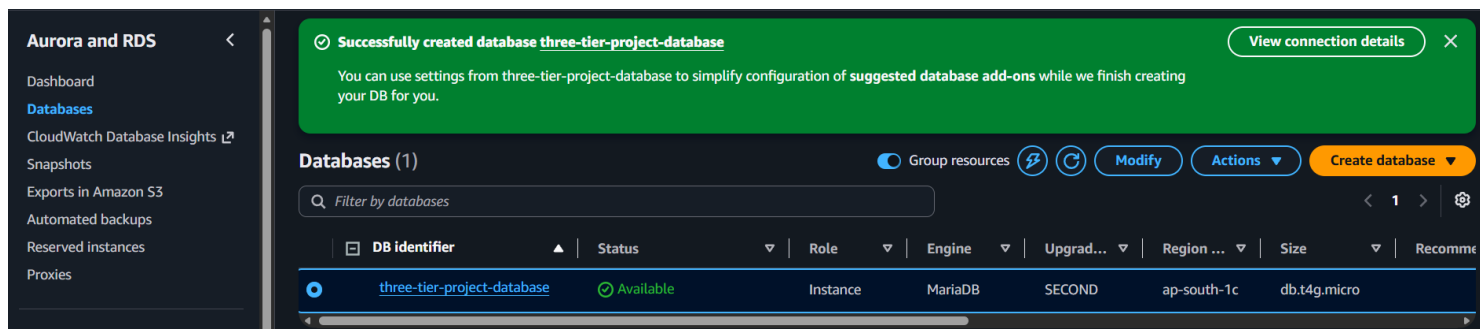
Step 18: Database Connection Error — Troubleshooting

Initial database connection failure captured before adding the JDBC connector/configuration.



Step 19: Amazon RDS Created

Amazon RDS database instance created for the database tier.



Step 20: MariaDB, RDS Connection & SQL Queries

MariaDB client verification, RDS connection and application SQL queries.

```
ec2-user@database:~  
[ec2-user@database ~]$ mariadb --version  
mariadb Ver 15.1 Distrib 10.5.29-MariaDB, for Linux (x86_64) using EditLine wrapper  
[ec2-user@database ~]$ curl -o global-bundle.pem https://truststore.pki.rds.amazonaws.com/global/global-bundle.pem  
% Total % Received % Xferd Average Speed Time Time Time Current  
Dload Upload Total Spent Left Speed  
100 165408 100 165408 0 0 213166 0 --:--:-- --:--:-- --:--:-- 213154  
[ec2-user@database ~]$ mysql -h three-tier-project-database.cha4a4assstm.ap-south-1.rds.amazonaws.com -P 3306 -u admin -p --ssl-verify-server-cert --ssl-ca=./global-bundle.pem  
Enter password:  
Welcome to the MariaDB monitor. Commands end with ; or \g.  
Your MariaDB connection id is 33  
Server version: 11.8.8-MariaDB-log managed by https://aws.amazon.com/rds/  
Copyright (C) 2000, 2018, Oracle, MariaDB Corporation Ab and others.  
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.  
MariaDB [(none)]> create table studentapp;  
ERROR 1046 (3D000): No database selected  
MariaDB [(none)]> create database studentapp;  
Query OK, 1 row affected (0.006 sec)  
MariaDB [(none)]> use studentapp;  
Database changed  
MariaDB [studentapp]> CREATE TABLE if not exists students(student_id INT NOT NULL AUTO_INCREMENT,  
-> student_name VARCHAR(100) NOT NULL,  
-> student_addr VARCHAR(100) NOT NULL,  
-> student_age VARCHAR(3) NOT NULL,  
-> student_qual VARCHAR(20) NOT NULL,  
-> student_percent VARCHAR(10) NOT NULL,  
-> student_year_passed VARCHAR(10) NOT NULL,  
-> PRIMARY KEY(student_id)  
-> );  
Query OK, 0 rows affected (0.016 sec)  
MariaDB [studentapp]> desc students;  
+-----+-----+-----+-----+-----+-----+  
| Field | Type | Null | Key | Default | Extra |  
+-----+-----+-----+-----+-----+-----+  
| student_id | int(11) | NO | PRI | NULL | auto_increment |  
| student_name | varchar(100) | NO | | NULL | |  
| student_addr | varchar(100) | NO | | NULL | |  
| student_age | varchar(3) | NO | | NULL | |  
| student_qual | varchar(20) | NO | | NULL | |  
| student_percent | varchar(10) | NO | | NULL | |  
| student_year_passed | varchar(10) | NO | | NULL | |  
+-----+-----+-----+-----+-----+-----+  
7 rows in set (0.010 sec)  
MariaDB [studentapp]> |
```

Step 21: MySQL Connector & Context Configuration

MySQL Connector/J added to the Tomcat library and context.xml opened for configuration.

```
root@application/opt/tomcat/conf
[ec2-user@application ~]$ sudo -i
[root@application ~]# cd /opt/tomcat/lib/
[root@application lib]# ls
annotations-api.jar  catalina-storeconfig.jar  el-api.jar  jsp-api.jar  tomcat-coyote.jar  tomcat-i18n-es.jar  tomcat-i18n-pt-BR.jar  tomcat-jni.jar  websocket-api.jar
catalina-ant.jar    catalina-tribes.jar       jasper-el.jar  servlet-api.jar  tomcat-dbc.jar  tomcat-i18n-fr.jar  tomcat-i18n-ru.jar  tomcat-util-scan.jar
catalina-ha.jar     catalina.jar              jasper.jar    tomcat-api.jar  tomcat-i18n-cs.jar  tomcat-i18n-ja.jar  tomcat-i18n-zh-CN.jar  tomcat-util.jar
catalina-ssi.jar    ecj-4.20.jar              jaspic-api.jar  tomcat-coyote-ffm.jar  tomcat-i18n-de.jar  tomcat-i18n-ko.jar  tomcat-jdbc.jar  tomcat-websocket.jar
[root@application lib]# curl -L -o https://github.com/Sushant123456789/three-tier-war-file/raw/refs/heads/main/mysql-connector.jar
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total   Spent    Left   Speed
0 0 0 0 0 0 0 0 0:00:02 0:00:02 --:--:-- 2191k
[root@application lib]# ls
annotations-api.jar  catalina-storeconfig.jar  el-api.jar  jsp-api.jar  tomcat-coyote-ffm.jar  tomcat-i18n-de.jar  tomcat-i18n-ko.jar  tomcat-jdbc.jar  tomcat-websocket.jar
catalina-ant.jar    catalina-tribes.jar       jasper-el.jar  mysql-connector.jar  tomcat-coyote.jar  tomcat-i18n-es.jar  tomcat-i18n-pt-BR.jar  tomcat-jni.jar  websocket-api.jar
catalina-ha.jar     catalina.jar              jasper.jar    servlet-api.jar  tomcat-dbc.jar  tomcat-i18n-fr.jar  tomcat-i18n-ru.jar  tomcat-util-scan.jar
catalina-ssi.jar    catalina.jar              jaspic-api.jar  tomcat-api.jar  tomcat-i18n-cs.jar  tomcat-i18n-ja.jar  tomcat-i18n-zh-CN.jar  tomcat-util.jar
[root@application lib]# cd
[root@application ~]# cd /opt/tomcat/conf/
[root@application conf]# ls
Catalina  catalina.policy  catalina.properties  context.xml  jaspic-providers.xml  jaspic-providers.xsd  logging.properties  server.xml  tomcat-users.xml  tomcat-users.xsd  web.xml
[root@application conf]# sudo vim context.xml
```

Step 22: Tomcat Context — RDS Database Configuration

Tomcat context configured with the RDS database connection details.

```
root@application/opt/tomcat/conf
<?xml version="1.0" encoding="UTF-8"?>
<!--
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contributor license agreements. See the NOTICE file distributed with
this work for additional information regarding copyright ownership.
The ASF licenses this file to You under the Apache License, Version 2.0
(the "License"); you may not use this file except in compliance with
the License. You may obtain a copy of the License at

    http://www.apache.org/licenses/LICENSE-2.0

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distributed under the License is distributed on an "AS IS" BASIS,
WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
See the License for the specific language governing permissions and
limitations under the License.
-->
<!-- The contents of this file will be loaded for each web application -->
<Context>

    <!-- Default set of monitored resources. If one of these changes, the    -->
    <!-- web application will be reloaded.                                -->
    <WatchedResource>WEB-INF/web.xml</WatchedResource>
    <WatchedResource>WEB-INF/tomcat-web.xml</WatchedResource>
    <WatchedResource>${catalina.base}/conf/web.xml</WatchedResource>

    <!-- Uncomment this to disable session persistence across Tomcat restarts -->
    <!--
    <Manager pathname="" />
    -->

    <Resource name="jdbc/TestDB" auth="Container" type="javax.sql.DataSource"
maxTotal="500" maxIdle="30" maxWaitMillis="1000"
username="admin" password="prgwoiWedr" driverClassName="com.mysql.jdbc.Driver"
url="jdbc:mysql://three-tier-project-database.cha4a4assstm.ap-south-1.rds.amazonaws.com:3306/studentapp?useUnicode=yes&amp;characterEncoding=utf8" />

</Context>
```

Step 23: Final Tomcat & Nginx Restart

Tomcat and Nginx restarted after database configuration.

```
ec2-user@proxy:~$ cd ../bin/
[root@application bin]# ls
bootstrap.jar  catalina.sh  commons-daemon-native.tar.gz  configtest.sh  digest.sh  setclasspath.bat  shutdown.sh  tomcat-juli.jar  tool-wrapper.sh
catalina-tasks.xml  ciphers.bat  commons-daemon.jar  daemon.sh  makebase.bat  setclasspath.sh  startup.bat  tomcat-native.tar.gz  version.bat
catalina.bat  ciphers.sh  configtest.bat  digest.bat  makebase.sh  shutdown.bat  startup.sh  tool-wrapper.bat  version.sh
[root@application bin]# ./catalina.sh stop
Using CATALINA_BASE:   /opt/tomcat
Using CATALINA_HOME:   /opt/tomcat
Using CATALINA_TMPDIR: /opt/tomcat/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /opt/tomcat/bin/bootstrap.jar:/opt/tomcat/bin/tomcat-juli.jar
Using CATALINA_OPTS:
NOTE: Picked up JDK_JAVA_OPTIONS:  --add-opens=java.base/java.lang=ALL-UNNAMED --add-opens=java.base/java.lang.invoke=ALL-UNNAMED --add-opens=java.base/java.lang.reflect=ALL-UNNAMED --add-opens=java.base/java.i
o=ALL-UNNAMED --add-opens=java.base/java.util=ALL-UNNAMED --add-opens=java.base/java.util.concurrent=ALL-UNNAMED --add-opens=java.rmi/sun.rmi.transport=ALL-UNNAMED
[root@application bin]# ./catalina.sh start
Using CATALINA_BASE:   /opt/tomcat
Using CATALINA_HOME:   /opt/tomcat
Using CATALINA_TMPDIR: /opt/tomcat/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /opt/tomcat/bin/bootstrap.jar:/opt/tomcat/bin/tomcat-juli.jar
Using CATALINA_OPTS:
Tomcat started.
[root@application bin]#
logout
[ec2-user@application ~]$
logout
Connection to 10.0.5.48 closed.
[ec2-user@proxy ~]$ sudo systemctl reload nginx
[ec2-user@proxy ~]$
```

Step 24: Final Student Form

Final student registration form displayed.

three-tier-project-database - Data | User Data

Not secure 13.207.41.4

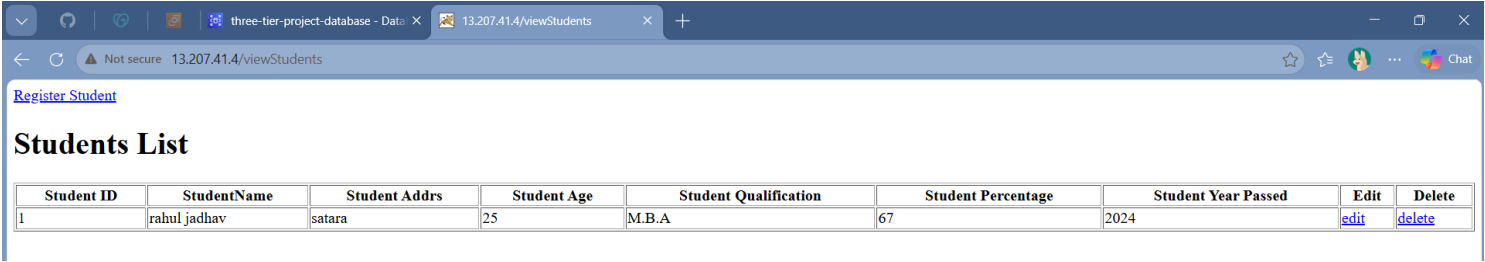
Student Registration Form

Student Name	rahul jadhav
Student Address	satara
Student Age	25
Student Qualification	M.B.A
Student Percentage	67
Year Passed	2024

register

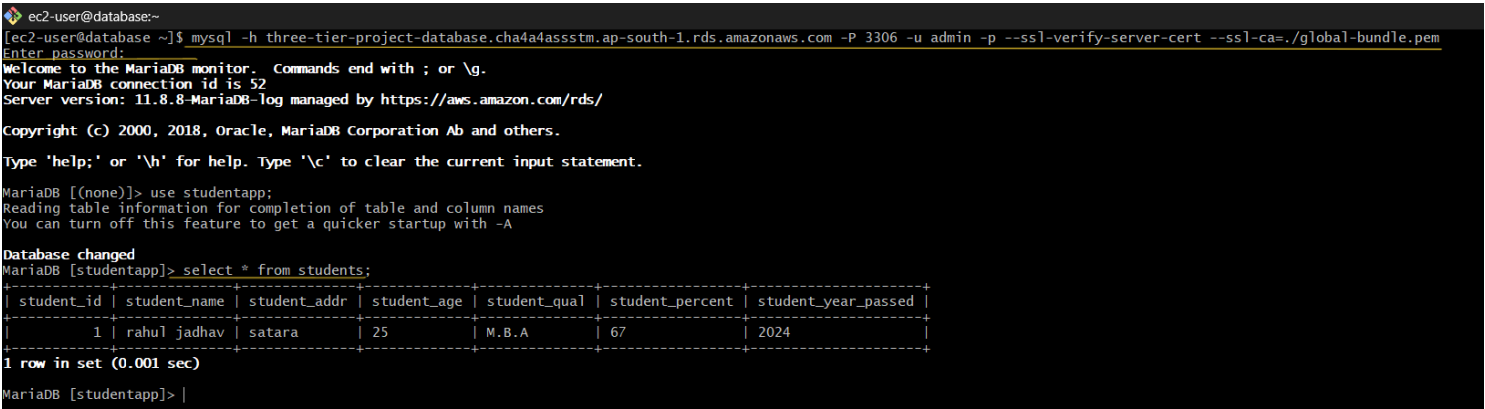
Step 25: Successful Student List Output

Successful application output showing the submitted student data.



Step 26: Database Data Verification

Database data verified after successful application submission.



CONCLUSION

3-Tier Java Application deployed on AWS

Project Summary

- Web Tier: Nginx on a public EC2 instance works as the reverse proxy.
- Application Tier: Apache Tomcat hosts the Java Student Management application in Private Subnet 1.
- Database Tier: Amazon RDS for MySQL stores the application data in Private Subnet 2.
- JDBC and MySQL Connector/J provide database connectivity from the Java application.
- The application was tested through the public Nginx endpoint and student data was successfully stored and verified in the database.

Final Request Flow

User → Internet → Nginx → Tomcat / Java Application → JDBC → RDS MySQL

Final Result

The three-tier Java application was successfully deployed, accessed through Nginx, connected to Amazon RDS, and verified with successfully stored student data.